CONVEX EAR DECOMPOSITION OF AUGMENTED BERGMAN COMPLEXES

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Editorial note. Original author statement, entire paired proof, definitions and supporting arguments retained. No added mathematical hypothesis, proof repair or correctness certification.

Editorial note. Full article context is retained to preserve internal references, prerequisite constructions, and native diagrams. Only this task primary statement is counted.

Editorial note. Mechanical rendering adaptation: removed the stray semicolon after native tikzstyle at source line 362; diagram content and geometry unchanged.

Allowed standard prerequisites (editorial).

- Standard graph, matroid and enumerative combinatorics. The source article definitions, cited standard results and bibliography are retained.. Original prerequisites; no new assumptions or author-proof replacements.

1. Introduction

Two well-studied objects that one may associate to a matroid M are the *independence complex*, denoted by I_{M}, and the *Bergman complex*, denoted by $\underline{\Delta}_{\mathsf{M}}$. Recently, these simplicial complexes have received considerable attention. Despite their simple definitions, several questions about their face enumeration have turned out to be very challenging. As a notable example, the recent breakthroughs by Brändén and Huh [BH20] and by Anari, Liu, Oveis-Gharan and Vinzant [ALOGV24] proved a long-standing conjecture by Mason, claiming that the entries of the f-vector of I_{M} form an *ultra log-concave* sequence. Moreover, the h-polynomial of the Bergman complex $\underline{\Delta}_{\mathsf{M}}$ was conjectured to be real-rooted (therefore, also ultra log-concave) in recent work of the first named author and Kalampogia-Evangelinou [AKE23]. The conjecture was verified there in interesting special cases, such as partition lattices and their standard type B analogues (also known as braid matroids of types A and B, respectively) and subspace lattices.

In the *singular* Hodge theory of matroids, developed by Braden, Huh, Matherne, Proudfoot and Wang [BHM$^+$22, BHM$^+$23], an instrumental role is played by a third simplicial complex, called the *augmented Bergman complex*. This simplicial complex 'interpolates' between I_{M} and $\underline{\Delta}_{\mathsf{M}}$ and contains both as subcomplexes. The focus of this paper is its face enumeration.

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Throughout this paper M will denote a matroid on the ground set $E = [n] := \{1, 2, \ldots, n\}$, with lattice of flats $\mathscr{L}(\mathsf{M})$. We let $U = \{u_i : i \in E\}$ and $V_{\mathsf{M}} = \{v_F : F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}\}$ be disjoint sets which are in one-to-one correspondence with E and $\mathscr{L}(\mathsf{M}) \smallsetminus \{E\}$, respectively.

Definition 1.1 The *augmented Bergman complex* of M, denoted by Δ_{M}, is the simplicial complex on the ground set $U \cup V_{\mathsf{M}}$ whose faces are the sets of the form

$$\{u_i : i \in S\} \cup \{v_{F_1}, v_{F_2}, \ldots, v_{F_k}\}, \tag{1}$$

where $S \in \mathrm{I}_{\mathsf{M}}$ is an independent set of M, the $F_1, F_2, \ldots, F_k \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}$ are proper flats and either $S \subseteq F_1 \subset F_2 \subset \cdots \subset F_k$ and $k \geq 1$, or $S \subseteq E$ and $k = 0$.

Augmented Bergman complexes were introduced in [BHM+22]. Their combinatorial and topological properties were studied in [BKR+22, Jef23], where augmented Bergman complexes were shown to be shellable and, in fact, vertex decomposable, respectively. In particular, augmented Bergman complexes are Cohen–Macaulay (over $\mathbb{Z}$ and any field). As a well known consequence of either shellability or Cohen–Macaulayness (see Sections II.3 and III.2 of [Sta96]), their h-vectors have nonnegative entries.

The main contribution of this paper is a new list of inequalities that the h-vector of Δ_{M} satisfies. The main technical tool employed is that of a convex ear decomposition (see Section 2.3), a property that has been established for the independence complex I_{M} by Chari [Cha97] and for the Bergman complex $\underline{\Delta}_{\mathsf{M}}$ by Nyman and Swartz [NS04]. This property, in addition to yielding inequalities for h-vectors, also establishes the double Cohen–Macaulayness of the complex: a simplicial complex Δ is said to be *doubly Cohen–Macaulay* if it is Cohen–Macaulay and removing any vertex results in a subcomplex that is Cohen–Macaulay of the same dimension as Δ (see [Sta96, Section III.3]).

Theorem 1.2 *The augmented Bergman complex Δ_{M} has a convex ear decomposition for every matroid M. In particular, Δ_{M} is doubly Cohen–Macaulay,*

$$h_0(\Delta_{\mathsf{M}}) \leq h_1(\Delta_{\mathsf{M}}) \leq \cdots \leq h_{\lfloor d/2 \rfloor}(\Delta_{\mathsf{M}}) \tag{2}$$

and

$$h_i(\Delta_{\mathsf{M}}) \leq h_{d-i}(\Delta_{\mathsf{M}}) \tag{3}$$

for $0 \leq i \leq d/2$, where d is the rank of M and $(h_i(\Delta_{\mathsf{M}}))_{0 \leq i \leq d}$ is the h-vector of Δ_{M}.

Inequalities (2) and (3), often referred to as the *top-heavy* inequalities for the h-vector of Δ_{M}, follow from the existence of a convex ear decomposition [Cha97]. They have been established for several fundamental enumerative invariants of matroids. For instance, as already mentioned, the independence complex (after removing coloops) and the Bergman complex of a matroid have convex ear decompositions. As a result, they have top-heavy (or strongly flawless, in other terminology) h-vectors and are doubly Cohen–Macaulay. The h-vector of the broken circuit complex, another simplicial complex of fundamental importance associated to a matroid, is top-heavy thanks to some deep results in the intersection theory of Chern–Schwartz–MacPherson cycles of matroids by Ardila, Denham and Huh [ADH23]; see also [JKL18] for an earlier partial result. Furthermore, the sequence of Whitney numbers

of the second kind of a matroid is top-heavy thanks to an important result by Braden, Huh, Matherne, Proudfoot and Wang [BHM+23, Theorem 1.1].

The content and structure of this paper is as follows. Theorem 1.2 is proven in Section 3; the proof may be viewed as an extension of the proof of the corresponding result of [NS04] for the Bergman complex. Section 2 explains relevant definitions and background on simplicial complexes, and matroid complexes in particular. Moreover, it includes a technical lemma (see Lemma 2.3) which is crucial for the proof of Theorem 1.2 and provides useful formulas for the f-vector and the h-vector of the augmented Bergman complex (see Propositions 2.7 and 2.9). Section 4 exhibits explicit counterexamples showing that top-heaviness cannot be upgraded to unimodality (or log-concavity). Section 5 proves an augmented analogue of a formula conjectured by Hameister, Rao and Simpson [HRS21] for the h-polynomial of the Bergman complex in the special case of uniform matroids.

2. Simplicial complexes and matroids

This section includes background material on simplicial complexes, especially those related to matroids, as well as some preliminary results, which are essential in understanding Theorem 1.2 and its proof. We will assume familiarity with basic notions about simplicial complexes and matroids and suggest [Bjö92, Oxl11, Sta96] as general references.

2.1. **Face enumeration.** Let Δ be a $(d-1)$-dimensional simplicial complex and $f_{i-1}(\Delta)$ be the number of $(i-1)$-dimensional faces of Δ, for $i \geq 0$. The sequence $(f_{-1}(\Delta), \ldots, f_{d-1}(\Delta))$ is called the *f-vector* of Δ. The *f-polynomial* of Δ is defined as

$$f(\Delta, x) = \sum_{F \in \Delta} x^{|F|} = \sum_{i=0}^{d} f_{i-1}(\Delta) x^i.$$

Here we follow the usual convention that $f_{-1}(\Delta) = 1$ whenever Δ is nonempty and $f_{-1}(\Delta) = 0$ otherwise.

Correspondingly, the *h-vector* $(h_0(\Delta), h_1(\Delta), \ldots, h_d(\Delta))$ and the *h-polynomial* $h(\Delta, x)$ of Δ are defined by a linear transformation of the f-vector:

$$\begin{aligned} h(\Delta, x) &= \sum_{F \in \Delta} x^{|F|}(1-x)^{d-|F|} \\ &= \sum_{i=0}^{d} f_{i-1}(\Delta)\, x^i (1-x)^{d-i} = \sum_{i=0}^{d} h_i(\Delta) x^i. \end{aligned} \tag{4}$$

Equivalently,

$$f(\Delta, x) = \sum_{i=0}^{d} h_i(\Delta)\, x^i\, (1+x)^{d-i}. \tag{5}$$

In other words, to obtain $f(\Delta, x)$ from $h(\Delta, x)$, one should first reverse (with respect to d) the coefficients of $h(\Delta, x)$, then substitute $x+1$ for x and finally reverse the order of the coefficients again.

Remark 2.1 Some sources (for instance, [ADH23, Bjö92]) define the f-polynomial and the h-polynomial with their coefficients in reverse order. Although there is no ambiguity in the meaning of $f_i(\Delta)$ and $h_i(\Delta)$, the way in which the coefficients appear does matter when considering some properties of the polynomials, such as ultra log-concavity.

Example 2.2 Let Δ be the one-dimensional simplicial complex consisting of $\{a, b\}$, $\{b, c\}$, $\{c, d\}$, $\{d, e\}$, $\{a, e\}$, $\{b, e\}$ and their subsets, where a, b, c, d, e are pairwise distinct (see Figure 1). Then, $f(\Delta, x) = 1 + 5x + 6x^2$ and hence $h(\Delta, x) = (1 - x)^2 + 5x(1 - x) + 6x^2 = 1 + 3x + 2x^2$ and $h(\Delta) = (1, 3, 2)$.

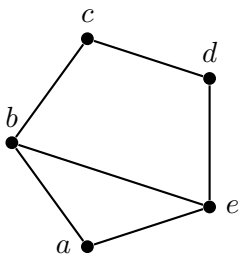


FIGURE 1. The simplicial complex of Example 2.2.

2.2. **Shellability.** Let Δ be a $(d-1)$-dimensional simplicial complex and let $\mathscr{F}$ be the set of facets (maximal with respect to inclusion faces) of Δ. Assume that Δ is pure, meaning that every facet has exactly d elements. Then, Δ is said to be *shellable* if $\mathscr{F}$ can be linearly ordered so that the following condition holds: Given $F, G \in \mathscr{F}$ with F preceding G in this order, there exists $F' \in \mathscr{F}$ which precedes G such that $F \cap G \subseteq F' \cap G$ and $|F' \cap G| = d-1$. Such a linear ordering is called a *shelling* of Δ. Shellability has strong consequences on the combinatorial and topological properties of Δ. For instance, it implies that Δ has a nonnegative h-vector and that it is Cohen–Macaulay (over $\mathbb{Z}$ and any field); see [Bjö84, Section 1] [Bjö92, Section 7.2] and [Sta96, Section III.2] for more information about this important concept.

Let $V = \{v_1, v_2, \ldots, v_d\}$ be a d-element set. Let sd_V denote the barycentric subdivision of the boundary complex of the simplex 2^V. The faces of sd_V are all chains of nonempty proper subsets of V. Consider another d-element set $U = \{u_1, u_2, \ldots, u_d\}$ which is disjoint from sd_V, and set $u_S := \{u_i : i \in S\}$ for $S \subseteq [d]$ and $v_S := \{v_i : i \in S\}$ for $S \subset [d]$.

Consider the simplicial complex Ω_V having vertex set $U \cup \{v_S : S \subset [d]\}$ and faces given by the sets of the form

$$u_S \cup \{v_{S_1}, v_{S_2}, \ldots, v_{S_k}\},$$

where $S \subseteq S_1 \subset S_2 \subset \cdots \subset S_k \subset [d]$ and $k \geq 1$, or $S \subseteq [d]$ and $k = 0$. The complex Ω_V is combinatorially isomorphic to the boundary complex of a d-dimensional polytope known as the simplicial stellohedron [BKR+22, Example 5] [PRW08, Section 10.4].

More generally, given a pure $(d-2)$-dimensional subcomplex Γ of sd_V, we define $\Omega_V(\Gamma)$ as the subcomplex of Ω_V whose facets are the facets F of Ω_V for which $F \setminus (U \cup \{v_\varnothing\})$ is a nonempty face of Γ. By definition, $\Omega_V(\Gamma)$ is a pure $(d-1)$-dimensional subcomplex of $\Omega_V \setminus \{U\}$ and $\Omega_V(\Gamma) = \Omega_V \setminus \{U\}$ if and only if $\Gamma = \partial(2^V)$.

Lemma 2.3 *For every d-element set V and every shellable $(d-2)$-dimensional subcomplex Γ of sd_V, the simplicial complex $\Omega_V(\Gamma)$ is a proper shellable $(d-1)$-dimensional subcomplex of Ω_V.*

Proof. Let $\mathscr{F}$ be the set of facets of $\Omega_V(\Gamma)$ and for $S \subset [d]$, let $\mathscr{F}_S$ be the set of those facets $F \in \mathscr{F}$ for which $F \cap U = u_S$.

Given $S \subset [d]$, we consider a maximal chain in the poset of nonempty subsets of S, ordered by inclusion, and let $\mathscr{C}_S \in \mathrm{sd}_V$ be the corresponding chain of proper subsets of V. Then, $\mathscr{F}_S$ is in one-to-one correspondence with the set of facets of the link $\mathrm{link}_\Gamma(\mathscr{C}_S)$, which is a shellable subcomplex of Γ of codimension equal to $|S|$. For example, $\mathscr{F}_\varnothing$ is in one-to-one correspondence with the set of facets of Γ and $\mathscr{F}_{\{u_i\}}$ is in one-to-one correspondence with the set of facets of $\mathrm{link}_\Gamma(v_i)$ for every $i \in [d]$.

Let us linearly order the facets of $\Omega_V(\Gamma)$ subject to the following conditions:

(a) For all $S, T \subset [d]$, the elements of $\mathscr{F}_S$ appear before those of $\mathscr{F}_T$ whenever $|S| < |T|$.
(b) For every $S \subset [d]$, the elements of $\mathscr{F}_S$ appear relative to each other according to a shelling order of $\mathrm{link}_\Gamma(\mathscr{C}_S)$.

We now show that such a linear order of $\mathscr{F}$ is a shelling of $\Omega_V(\Gamma)$. Given $F, G \in \mathscr{F}$ with F preceding G, we need to verify that there exists $F' \in \mathscr{F}$ which precedes G such that $F \cap G \subseteq F' \cap G$ and $|F' \cap G| = d-1$. Indeed, this follows from condition (b) if $F, G \in \mathscr{F}_S$ for some $S \subset [d]$. Otherwise, we have $F \in \mathscr{F}_S$ and $G \in \mathscr{F}_T$ for some $S, T \subset [d]$ with $|S| \leq |T|$ and $S \neq T$. Then, there exists $i \in (T \smallsetminus S)$ and the set obtained from G by removing u_i and including $v_{T \smallsetminus \{i\}}$ (which is equal to $v_\varnothing$, if $T = \{i\}$) can serve as the desired facet F'. □

Example 2.4 (a) For $d = 2$, Ω_V is the one-dimensional simplicial complex having facets $\{v_\varnothing, v_1\}$, $\{v_\varnothing, v_2\}$, $\{u_1, v_1\}$, $\{u_2, v_2\}$ and $\{u_1, u_2\}$, where we have written v_i instead of $v_{\{i\}}$ for $i \in \{1, 2\}$.

(b) Adopting similar notation for v_S, let $d = 3$ and Γ be the subcomplex of sd_V with facets $\{v_1, v_{12}\}$ and $\{v_2, v_{12}\}$. Then, $\Omega_V(\Gamma)$ has facets $\{v_\varnothing, v_1, v_{12}\}$, $\{v_\varnothing, v_2, v_{12}\}$, $\{u_1, v_1, v_{12}\}$, $\{u_2, v_2, v_{12}\}$ and $\{u_1, u_2, v_{12}\}$. Any linear ordering of these facets which lists $\{v_\varnothing, v_1, v_{12}\}$ and $\{v_\varnothing, v_2, v_{12}\}$ first, in any relative order, followed by $\{u_1, v_1, v_{12}\}$ and $\{u_2, v_2, v_{12}\}$, in any relative order, followed by $\{u_1, u_2, v_{12}\}$ is a shelling of $\Omega_V(\Gamma)$ which satisfies the conditions in the proof of Lemma 2.3.

2.3. **Convex ear decompositions.** Let Δ be a $(d-1)$-dimensional simplicial complex. A *convex ear decomposition* of Δ is a sequence $(\Delta_1, \Delta_2, \ldots, \Delta_m)$ of subcomplexes of Δ such that:

(i) $\Delta = \bigcup_{i=1}^m \Delta_i$,
(ii) Δ_1 is isomorphic to the boundary complex of a d-dimensional simplicial polytope and, for $2 \leq k \leq m$, Δ_k is a $(d-1)$-dimensional simplicial ball which is isomorphic to a proper subcomplex of the boundary complex of a d-dimensional simplicial polytope; and
(iii) $\Delta_k \cap \left(\bigcup_{i=1}^{k-1} \Delta_i\right) = \partial \Delta_k$ for $2 \leq k \leq m$.

Convex ear decompositions were introduced by Chari [Cha97] as a tool to prove the top-heavy inequalities for h-vectors of simplicial complexes [Cha97, Corollary 2]. Swartz [Swa06, Theorem 4.1] showed that every simplicial complex Δ which admits a convex ear decomposition is doubly Cohen–Macaulay, meaning that Δ is Cohen–Macaulay and that every subcomplex which is obtained by removing a vertex from Δ is Cohen–Macaulay, of the same dimension as Δ. We refer to [Sta96, Section III.3] for more information about this important class of complexes.

Example 2.5 Let Δ be as in Example 2.2. There is the convex ear decomposition (Δ_1, Δ_2) of Δ, where Δ_1 is the one-dimensional sphere with facets $\{b, c\}$, $\{c, d\}$, $\{d, e\}$ and $\{b, e\}$ and Δ_2 is the one-dimensional ball with facets $\{a, b\}$ and $\{a, e\}$. Alternatively, one can choose Δ_1 to have all facets of Δ other than $\{b, e\}$ and Δ_2 to have the single facet $\{b, e\}$.

Remark 2.6 Let $\Sigma_1, \Sigma_2, \ldots, \Sigma_m$ be subcomplexes of a $(d-1)$-dimensional simplicial complex Δ such that:

(i) $\Delta = \bigcup_{i=1}^{m} \Sigma_i$,
(ii) Σ_i is isomorphic to the boundary complex of a d-dimensional simplicial polytope for every $i \in [m]$; and
(iii) $\Sigma_k \cap \left(\bigcup_{i=1}^{k-1} \Sigma_i\right)$ is a $(d-1)$-dimensional simplicial ball, for all $2 \leq k \leq m$.

Then, Δ admits a convex ear decomposition in which the role of Δ_1 is played by Σ_1 and the role of Δ_k for $2 \leq k \leq m$ is played by the complex obtained from Σ_k by removing the interior faces of the ball $\Sigma_k \cap \left(\bigcup_{i=1}^{k-1} \Sigma_i\right)$.

2.4. **Matroids.** Let M be a matroid on the ground set E, having lattice of flats $\mathscr{L}(\mathsf{M})$. The *Bergman complex* of M, denoted by $\underline{\Delta}_{\mathsf{M}}$, is defined as the order complex of the poset $\mathscr{L}(\mathsf{M}) \smallsetminus \{\widehat{0}, \widehat{1}\}$, meaning the simplicial complex consisting of all chains in that poset. Note that $\underline{\Delta}_{\mathsf{M}} = \{\varnothing\}$ if M has rank zero or one.

The following proposition allows one to compute the f-polynomial of the augmented Bergman complex in terms of the f-polynomials of Bergman complexes of contractions of M.

Proposition 2.7 *Let M be a matroid with set of bases $\mathcal{B}(\mathsf{M})$. Then, the f-polynomial of the augmented Bergman complex of M is given by the formula*

$$f(\Delta_{\mathsf{M}}, x) = |\mathcal{B}(\mathsf{M})|\, x^d + (x+1) \sum_{\substack{S \in \mathrm{I}_{\mathsf{M}} \\ S \notin \mathcal{B}(\mathsf{M})}} x^{|S|}\, f(\underline{\Delta}_{\mathsf{M}/S}, x),$$

where d is the rank of M and M/S is the contraction of M by S.

Proof. By definition, a face of Δ_{M} has the form $\{u_i : i \in S\} \cup \{v_{F_1}, \ldots, v_{F_k}\}$, where S is an independent set and $F_1 \subset \cdots \subset F_k$ is a (possibly empty) chain of proper flats containing S. After fixing the independent set S, we have to enumerate chains of flats in the contraction M/S, but allowing the first element in the chain to be equal to the bottom element while disallowing the last element of the chain to be equal to the top element. In other words, we

are enumerating the faces of a cone over the Bergman complex $\underline{\Delta}_{\mathsf{M}/S}$. There is one special case in which the operation is not an actual coning: when $\widehat{0} = \widehat{1}$ in the poset $\mathscr{L}(\mathsf{M}/S)$, which happens if and only if M/S has rank zero or, equivalently, when S is a basis of M. Summing over all independent sets yields the formula of the statement. □

Example 2.8 Consider the uniform matroid $\mathsf{U}_{d,n}$ of rank d on n elements. The preceding formula yields

$$f(\Delta_{\mathsf{U}_{d,n}}, x) = \binom{n}{d} x^d + (x+1) \sum_{j=0}^{d-1} \binom{n}{j} x^j \, f(\underline{\Delta}_{\mathsf{U}_{d-j,n-j}}, x).$$

Notice that the expression on the right depends on the f-polynomials of Bergman complexes of uniform matroids. These can be computed, for example, using [FS24, Proposition 8.10]. In Section 5 we will establish a surprising connection between $f(\Delta_{\mathsf{U}_{d,n}}, x)$ and the Hilbert–Poincaré series of augmented Chow rings of uniform matroids. □

The next result provides an alternative concrete formula for computing the h-polynomial of the augmented Bergman complex of a matroid M. Specifically, it is given by a convolution of h-polynomials of independence complexes of restrictions of M and Bergman complexes of contractions of M by flats (as opposed to independent sets).

Proposition 2.9 *The h-polynomial of the augmented Bergman complex of a matroid M on the ground set E is given by the formula*

$$h(\Delta_{\mathsf{M}}, x) = h(\mathrm{I}_{\mathsf{M}}, x) + x \sum_{F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}} h(\mathrm{I}_{\mathsf{M}|_F}, x) h(\underline{\Delta}_{\mathsf{M}/F}, x) \tag{6}$$

where, for $F \in \mathscr{L}(\mathsf{M})$, the matroids $\mathsf{M}|_F$ and M/F are the restriction of M to F and the contraction of M by F, respectively.

Proof. Given a flat $F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}$, let us denote by $\mathscr{G}_F$ the set of faces (1) of Δ_{M} for which $k \geq 1$ and $F_1 = F$. Then, Δ_{M} is the disjoint union of I_{M} and the sets $\mathscr{G}_F$. Therefore, by equation (4),

$$\begin{aligned} h(\Delta_{\mathsf{M}}, x) &= \sum_{G \in \mathrm{I}_{\mathsf{M}}} x^{|G|}(1-x)^{d-|G|} + \sum_{F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}} \sum_{G \in \mathscr{G}_F} x^{|G|}(1-x)^{d-|G|} \\ &= h(\mathrm{I}_{\mathsf{M}}, x) + \sum_{F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}} \sum_{G \in \mathscr{G}_F} x^{|G|}(1-x)^{d-|G|}. \end{aligned}$$

Since the sets $\{u_i : i \in S\}$ with $S \subseteq F$ are exactly the faces of $\mathrm{I}_{\mathsf{M}|_F}$ and the chains of flats $F_2 \subset \cdots \subset F_k \subset E$ with $F \subset F_2$ are exactly the faces of $\underline{\Delta}_{\mathsf{M}/F}$, we conclude that

$$\begin{aligned} \sum_{G \in \mathscr{G}_F} x^{|G|}(1-x)^{d-|G|} &= x \sum_{G' \in \mathrm{I}_{\mathsf{M}|_F}} x^{|G'|}(1-x)^{\mathrm{rk}(F)-|G'|} \sum_{G'' \in \underline{\Delta}_{\mathsf{M}/F}} x^{|G''|}(1-x)^{d-1-\mathrm{rk}(F)-|G''|} \\ &= x \sum_{F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}} h(\mathrm{I}_{\mathsf{M}|_F}, x) h(\underline{\Delta}_{\mathsf{M}/F}, x) \end{aligned}$$

for every $F \in \mathscr{L}(\mathsf{M}) \smallsetminus \{E\}$ and the proof follows. □

Example 2.10 For the uniform matroid $\mathsf{U}_{d,n}$ of rank d on n elements, equation (6) yields that

$$h(\Delta_{\mathsf{U}_{d,n}}, x) = \sum_{k=0}^{d} \binom{n-d+k-1}{d} x^k + x \sum_{k=0}^{d-1} \binom{n}{k} h(\underline{\Delta}_{\mathsf{U}_{d-k,n-k}}, x)$$
$$= \sum_{k=0}^{d} \binom{n-d+k-1}{d} x^k + x \sum_{k=0}^{d-1} \binom{n}{k} \sum_{w \in \mathfrak{S}_{n-k} \colon \mathrm{Des}(w) \subseteq [d-k-1]} x^{\mathrm{des}(w)},$$

where $\mathrm{Des}(w)$ and $\mathrm{des}(w)$ stand for the set and number of descents of a permutation w, respectively; see [ADK23, Example 2.2] for the justification of the last equality. This formula agrees with the fact that $h(\Delta_{\mathsf{U}_{n,n}}, x)$ is equal to the nth binomial Eulerian polynomial [BKR+22, Example 5] [PRW08, Section 10.4].

3. Proof of Theorem 1.2

This section proves Theorem 1.2. Let M be a matroid of rank d on the ground set $E = [n]$, as in the introduction.

Let us briefly recapitulate some notions that are needed for the proof of Theorem 1.2, following [Björ92, Section 7.3] and [NS04, Section 4].

Fix the total order $1 < 2 < \cdots < n$ of the ground set $E = [n]$ of M. Let us agree to write every d-subset $S = \{i_1, \ldots, i_d\} \subseteq E$ as an ordered d-tuple $[i_1, \ldots, i_d]$, where $i_1 < \cdots < i_d$. This induces the lexicographic order on the set of all d-subsets of E: we say that $[i_1, \ldots, i_d]$ *lexicographically precedes* $[j_1, \ldots, j_d]$ if there is an index ℓ such that $i_\ell < j_\ell$ and $i_m = j_m$ for all $m < \ell$. A *broken circuit* of M is a circuit of M with its minimum element removed. An *nbc basis* of M is a basis of M that does not contain a broken circuit; observe that all such bases must contain the element 1.

Consider a facet $\mathcal{F}$ of the order complex of $\mathscr{L}(\mathsf{M})$ or, in other words, a maximal chain of flats of M. All such chains are of the form $\mathcal{F} = \{\widehat{0} = F_0 \subset F_1 \subset \cdots \subset F_{d-1} \subset F_d = \widehat{1}\}$ and one may consider the ordered d-tuple $(a_1, \ldots, a_d)$, where $a_i = \min\{e \in E : \overline{F_{i-1} \cup \{e\}} = F_i\}$ for each $i \in [d]$. This d-tuple is called the *minimal labeling* associated to $\mathcal{F}$. It is evident that $\{a_1, \ldots, a_d\}$ is a basis of M. For $S \subseteq E$, we denote by $\overline{S}$ the flat of M spanned by S.

Given a basis B of M, we denote by sd_B the simplicial complex with faces

$$\{v_{\overline{S_1}}, v_{\overline{S_2}}, \ldots, v_{\overline{S_k}}\},$$

where $\varnothing \subset S_1 \subset S_2 \subset \cdots \subset S_k \subset B$. Clearly, sd_B is a subcomplex of $\underline{\Delta}_{\mathsf{M}}$ which is combinatorially isomorphic to sd_V for any d-element set V. Let B be a basis of M. Every facet of sd_B or, equivalently, every chain $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_{d-1} \subset S_d = B$, corresponds to a permutation $(a_1, \ldots, a_d)$ of B, where $S_i = S_{i-1} \cup \{a_i\}$ for $i \in [d]$.

The following result is obtained by relying on the techniques of Nyman and Swartz [NS04, Section 4].

Proposition 3.1 *Let $B_1, B_2, \ldots, B_m$ be the bases of M, sorted in the lexicographic order. Consider the subcomplex*

$$\Gamma_k = \mathrm{sd}_{B_k} \cap \left(\bigcup_{i<k} \mathrm{sd}_{B_i} \right) \tag{7}$$

of sd_{B_k} for $2 \leq k \leq m$.

(a) *The complex Γ_k is a shellable $(d-2)$-dimensional ball, if B_k is an nbc basis.*
(b) *Otherwise, $\Gamma_k = \mathrm{sd}_{B_k}$.*

Proof. Suppose first that B_k is *not* nbc. Consider any facet $\mathcal{F}$ of sd_{B_k}, say with corresponding permutation $(a_1, a_2, \ldots, a_d)$ of B_k, and let $j \in [d]$ be the smallest index for which $\{a_1, a_2, \ldots, a_j\}$ does contain a broken circuit. Replacing a_j with the smallest element of that circuit yields a permutation of a basis B_i with $i < k$ which also represents $\mathcal{F}$. This argument shows that $\mathcal{F}$ is a facet of sd_{B_i} and verifies part (b).

For $k \in [m]$, let Λ_k be the subcomplex of sd_{B_k} whose facets are the maximal chains

$$\{v_{\overline{S_1}}, v_{\overline{S_2}}, \ldots, v_{\overline{S_{d-1}}}\}$$

of $\mathscr{L}(\mathsf{M})$, where $\varnothing \subset S_1 \subset S_2 \subset \cdots \subset S_{d-1} \subset B_k$, whose minimal label coincides with the corresponding permutation of B_k. We claim that

$$\bigcup_{i=1}^{k} \mathrm{sd}_{B_i} = \bigcup_{i=1}^{k} \Lambda_i \tag{8}$$

for every $k \in [m]$. Indeed, the $\supseteq$ inclusion is trivial. For its reverse, let $\mathcal{F}$ be a facet of sd_{B_k} which does not belong to Λ_k and let $(a_1, a_2, \ldots, a_d)$ be the corresponding permutation of B_k. Since this permutation is not equal to the minimal labeling of the maximal chain of $\mathscr{L}(\mathsf{M})$ corresponding to $\mathcal{F}$, there exists an index $j \in [d]$ and an element $a_0 \in E$ with $a_0 < a_j$ such that

$$\overline{\{a_1, \ldots, a_{j-1}, a_j\}} = \overline{\{a_1, \ldots, a_{j-1}, a_0\}}.$$

Replacing a_j with a_0 yields a permutation of a basis B_i with $i < k$ which also represents $\mathcal{F}$. This shows that $k \geq 2$ and that $\mathcal{F}$ is a facet of $\bigcup_{i=1}^{k-1} \mathrm{sd}_{B_i}$. Hence, the claim follows by induction on k.

Suppose now that $2 \leq k \leq m$ and that B_k is nbc. It was shown in [NS04, Section 4] (see Propositions 4.4 and 4.5 there) that Λ_k is a $(d-2)$-dimensional shellable ball and that

$$\Lambda_k \cap \left(\bigcup_{i<k} \Lambda_i \right) = \partial \Lambda_k. \tag{9}$$

Combining equations (7), (8) and (9) shows that Γ_k is disjoint from the interior of the ball Λ_k. On the other hand, every facet of sd_{B_k} which does not belong to Λ_k must belong to the union $\bigcup_{i<k} \Lambda_i = \bigcup_{i<k} \mathrm{sd}_{B_i}$ and hence to Γ_k. It follows that Γ_k is the complementary ball to Λ_k within the sphere sd_{B_k}. Moreover, [NS04, Lemma 4.3] implies that the facets of Γ_k correspond to a set of permutations of B_k which forms an order ideal in the weak order on the group of permutations of B_k and hence Γ_k is also shellable by [Bjö84, Theorem 2.1]. □

Given a basis B of M, we denote by Ω_B the simplicial complex with faces

$$\{u_i : i \in S\} \cup \{v_{\overline{S_1}}, v_{\overline{S_2}}, \ldots, v_{\overline{S_k}}\},$$

where $S \subseteq S_1 \subset S_2 \subset \cdots \subset S_k \subset B$ and $k \geq 1$, or $S \subseteq B$ and $k = 0$. Clearly, Ω_B is a subcomplex of Δ_{M} which is combinatorially isomorphic to Ω_V for any d-element set V.

Proof of Theorem 1.2. Let $B_1, B_2, \ldots, B_m$ be the bases of M, sorted in the lexicographic order. Then, the augmented Bergman complex Δ_{M} is the union of its subcomplexes Ω_{B_i} for $i \in [m]$, each of which is combinatorially isomorphic to the boundary complex of the d-dimensional simplicial stellohedron. By the relevant definitions, for $2 \leq k \leq m$ we have

$$\Omega_{B_k} \cap \left(\bigcup_{i<k} \Omega_{B_i} \right) = \Omega_{B_k}(\Gamma_k), \tag{10}$$

where Γ_k is as in Proposition 3.1. Combined with that, Lemma 2.3 implies that $\Omega_{B_k}(\Gamma_k)$ is a (shellable) $(d-1)$-dimensional ball. Hence, by Remark 2.6, the sequence $(\Omega_{B_1}, \Omega_{B_2}, \ldots, \Omega_{B_m})$ of polytopal simplicial spheres induces a convex ear decomposition of Δ_{M}. □

4. On unimodality and log-concavity

This section shows that f- and h-vectors of augmented Bergman complexes are not well behaved with respect to unimodality and log-concavity. Let $p(x) = a_0 + a_1 x + \cdots + a_d x^d$ be a polynomial of degree at most d with nonnegative coefficients. We recall that $p(x)$ is called

- *unimodal*, if $a_0 \leq a_1 \leq \cdots \leq a_k \geq a_{k+1} \geq \cdots \geq a_d$ for some $k \in \{0, 1, \ldots, d\}$,
- *log-concave*, if $a_i^2 \geq a_{i-1} a_{i+1}$ for every $i \in [d-1]$,
- *ultra log-concave*, of order $m \geq d$, if

$$\frac{a_i^2}{\binom{m}{i}^2} \geq \frac{a_{i-1}}{\binom{m}{i-1}} \cdot \frac{a_{i+1}}{\binom{m}{i+1}}$$

  for every $i \in [d-1]$,
- *real-rooted*, if every root of $p(x)$ is real, or $p(x) \equiv 0$.

This is a hierarchy of increasingly stronger properties, where the implication log-concave $\Longrightarrow$ unimodal requires that $p(x)$ has no internal zeros, meaning there are no indices $i < j < k$ such that $a_i a_k \neq 0$ and $a_j = 0$. We refer to surveys by Stanley [Sta89] and Brändén [Brä15] for more information about unimodality, log-concavity and real-rootedness. We also point to the work of Liggett [Lig97] and Brändén and Huh [BH20] for ultra log-concavity and multivariate generalizations.

Let M be a matroid on n elements. The independence complex I_{M} was shown to have an ultra log-concave f-polynomial of order n in [BH20, ALOGV24] and a log-concave h-polynomial in [AHK18]. A stronger property has been conjectured for the h-polynomial of the Bergman complex.

Conjecture 4.1 ([AKE23]) The f-polynomial (equivalently, the h-polynomial) of $\underline{\Delta}_{\mathsf{M}}$ is real-rooted for every matroid M.

This conjecture, if true, implies that both f- and h-polynomials of Bergman complexes are ultra log-concave of any order, and in particular log-concave. These weaker properties are also open.

One may suspect that f- and h-polynomials of augmented Bergman complexes also satisfy some version of log-concavity. The following example rules out that possibility.

Example 4.2 Consider the uniform matroid $\mathsf{U}_{4,189}$. The f-vector of its augmented Bergman complex can be computed explicitly via Example 2.8:

$$f(\Delta_{\mathsf{U}_{4,189}}) = (1, 1125559, 11181114, 27774180, 69213375).$$

This f-vector is not log-concave, because

$$27774180^2 = 771405074672400 < 773882636199750 = 11181114 \cdot 69213375.$$

This example implies that h-vectors of augmented Bergman complexes are not always log-concave. In fact, the situation is even worse, as these h-vectors can even fail to be unimodal. For the uniform matroid $\mathsf{U}_{5,83}$ we have

$$h(\Delta_{\mathsf{U}_{5,83}}) = (1, 1933066, 28121900, 60710014, 28680319, 29034396).$$

The reader may observe that $60710014 > 28680319 < 29034396$. We have not been able to find a non-unimodal f-vector. □

Remark 4.3 We have shown that Δ_{M} is doubly Cohen–Macaulay. The first named author and Tzanaki have posed the question [AT21, Question 7.2] whether the h-vector of any $(d-1)$-dimensional doubly Cohen–Macaulay simplicial complex Δ satisfies the inequalities

$$\frac{h_0(\Delta)}{h_d(\Delta)} \leq \frac{h_1(\Delta)}{h_{d-1}(\Delta)} \leq \frac{h_2(\Delta)}{h_{d-2}(\Delta)} \leq \cdots \leq \frac{h_d(\Delta)}{h_0(\Delta)}. \tag{11}$$

These inequalities imply that $h_i(\Delta) \leq h_{d-i}(\Delta)$ for $0 \leq i \leq d/2$. We have not found any counterexamples among h-vectors of augmented Bergman complexes.

5. A connection to augmented Chow polynomials

Hameister, Rao and Simpson conjectured a surprising relation between the h-vectors of Bergman complexes and the Hilbert–Poincaré series of Chow rings of uniform matroids [HRS21, Conjecture 6.2]. Their conjecture was proved recently by Ferroni, Matherne, Stevens and Vecchi [FMSV24, Corollary 3.17]. Specifically, they established the following formula.

Theorem 5.1 ([FMSV24]) *The h-polynomial of the Bergman complex of $\mathsf{U}_{d,n}$ can be expressed as*

$$x^{d-1} h(\underline{\Delta}_{\mathsf{U}_{d,n}}, x^{-1}) = \sum_{i=1}^{d} \binom{n-i-1}{d-i} \underline{\mathrm{H}}_{\mathsf{U}_{i,n}}(x),$$

where $\underline{\mathrm{H}}_{\mathsf{M}}(x)$ stands for the Hilbert–Poincaré series of the Chow ring of M.

This section provides an analogue of this theorem for augmented Bergman complexes. We emphasize that, a priori, there are no reasons to expect such a concise relation between

these two classes of polynomials and it is surprising that it continues to hold true in the augmented setting.

We assume familiarity with basic notions about Chow rings of matroids and their Hilbert–Poincaré series; our notation here follows [FMSV24] closely, and we refer to that paper for any undefined terminology. One important distinction between the Chow polynomial and the augmented Chow polynomial is their definition on matroids with loops. In general, the Chow polynomial is defined to be zero if the matroid has a loop, whereas the augmented Chow polynomial of a matroid with loops does not change if one removes the loops.

Theorem 5.2 *The h-vector of the augmented Bergman complex of $\mathsf{U}_{d,n}$ can be expressed as*

$$x^d h(\Delta_{\mathsf{U}_{d,n}}, x^{-1}) = \sum_{i=0}^{d} \binom{n-i-1}{d-i} \mathrm{H}_{\mathsf{U}_{i,n}}(x),$$

where $\mathrm{H}_{\mathsf{M}}(x)$ stands for the Hilbert–Poincaré series of the augmented Chow ring of M.

Proof. We will employ the following formula, established in [FMSV24, Theorem 1.3 (iii)]:

$$\mathrm{H}_{\mathsf{M}}(x) = \sum_{F \in \mathscr{L}(\mathsf{M})} x^{\mathrm{rk}(F)}\, \underline{\mathrm{H}}_{\mathsf{M}/F}(x).$$

By applying this identity to uniform matroids, we have the following chain of equalities:

$$\begin{aligned}\sum_{i=0}^{d} \binom{n-i-1}{d-i} \mathrm{H}_{\mathsf{U}_{i,n}}(x) &= \sum_{i=0}^{d} \binom{n-i-1}{d-i} \left(x^i + \sum_{j=0}^{i-1} \binom{n}{j} x^j\, \underline{\mathrm{H}}_{\mathsf{U}_{i-j,n-j}}(x) \right) \\ &= \sum_{i=0}^{d} \binom{n-i-1}{d-i} x^i + \sum_{i=0}^{d} \sum_{j=0}^{i-1} \binom{n-i-1}{d-i} \binom{n}{j} x^j\, \underline{\mathrm{H}}_{\mathsf{U}_{i-j,n-j}}.\end{aligned}$$

Changing the order of summation in the second term and reindexing, we get

$$\begin{aligned}\sum_{i=0}^{d} \binom{n-i-1}{d-i} \mathrm{H}_{\mathsf{U}_{i,n}}(x) &= \sum_{i=0}^{d} \binom{n-i-1}{d-i} x^i + \sum_{j=0}^{d-1} \binom{n}{j} x^j \sum_{i=j+1}^{d} \binom{n-i-1}{d-i}\, \underline{\mathrm{H}}_{\mathsf{U}_{i-j,n-j}} \\ &= \sum_{i=0}^{d} \binom{n-i-1}{d-i} x^i + \sum_{j=0}^{d-1} \binom{n}{j} x^j \sum_{i=1}^{d-j} \binom{n-j-i-1}{d-j-i}\, \underline{\mathrm{H}}_{\mathsf{U}_{i,n-j}}.\end{aligned}$$

We now apply the formula of Theorem 5.1 to get

$$\begin{aligned}\sum_{i=0}^{d} \binom{n-i-1}{d-i} \mathrm{H}_{\mathsf{U}_{i,n}}(x) &= \sum_{i=0}^{d} \binom{n-i-1}{d-i} x^i + \sum_{j=0}^{d-1} \binom{n}{j} x^j \left(x^{d-j-1} h(\underline{\Delta}_{\mathsf{U}_{d-j,n-j}}, x^{-1}) \right) \\ &= \sum_{i=0}^{d} \binom{n-i-1}{d-i} x^i + \sum_{j=0}^{d-1} \binom{n}{j} x^{d-1} h(\underline{\Delta}_{\mathsf{U}_{d-j,n-j}}, x^{-1}).\end{aligned}$$

Finally, we reverse the order of the coefficients and compare to the identity of Example 2.10 and the proof follows. □

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References

[ADH23] Federico Ardila, Graham Denham, and June Huh, *Lagrangian geometry of matroids*, J. Amer. Math. Soc. **36** (2023), no. 3, 727–794. MR 4583774

[ADK23] Christos A. Athanasiadis, Theo Douvropoulos, and Katerina Kalampogia-Evangelinou, *Two classes of posets with real-rooted chain polynomials*, Electron. J. Combin. **31** (2024), no. 4, Paper No. 4.16, 22. MR 4816497

[AHK18] Karim Adiprasito, June Huh, and Eric Katz, *Hodge theory for combinatorial geometries*, Ann. of Math. (2) **188** (2018), no. 2, 381–452. MR 3862944

[AKE23] Christos A. Athanasiadis and Katerina Kalampogia-Evangelinou, *Chain enumeration, partition lattices and polynomials with only real roots*, Comb. Theory **3** (2023), no. 1, Paper No. 12, 21. MR 4565299

[ALOGV24] Nima Anari, Kuikui Liu, Shayan Oveis Gharan, and Cynthia Vinzant, *Log-concave polynomials II: High-dimensional walks and an FPRAS for counting bases of a matroid*, Ann. of Math. (2) **199** (2024), no. 1, 259–299. MR 4681146

[AT21] Christos A. Athanasiadis and Eleni Tzanaki, *Symmetric decompositions, triangulations and real-rootedness*, Mathematika **67** (2021), no. 4, 840–859. MR 4304414

[BH20] Petter Brändén and June Huh, *Lorentzian polynomials*, Ann. of Math. (2) **192** (2020), no. 3, 821–891. MR 4172622

[BHM+22] Tom Braden, June Huh, Jacob P. Matherne, Nicholas Proudfoot, and Botong Wang, *A semi-small decomposition of the Chow ring of a matroid*, Adv. Math. **409** (2022), Paper No. 108646, 49. MR 4477425

[BHM+23] Tom Braden, June Huh, Jacob P. Matherne, Nicholas Proudfoot, and Botong Wang, *Singular Hodge theory for combinatorial geometries*, arXiv e-prints (2023), arXiv:2010.06088v4.

[Bjö84] Anders Björner, *Some combinatorial and algebraic properties of Coxeter complexes and Tits buildings*, Adv. in Math. **52** (1984), no. 3, 173–212. MR 744856

[Bjö92] ———, *The homology and shellability of matroids and geometric lattices*, Matroid applications, Encyclopedia Math. Appl., vol. 40, Cambridge Univ. Press, Cambridge, 1992, pp. 226–283. MR 1165544

[BKR+22] Elisabeth Bullock, Aidan Kelley, Victor Reiner, Kevin Ren, Gahl Shemy, Dawei Shen, Brian Sun, and Zhichun Joy Zhang, *Topology of augmented Bergman complexes*, Electron. J. Combin. **29** (2022), no. 1, Paper No. 1.31, 19. MR 4396472

[Brä15] Petter Brändén, *Unimodality, log-concavity, real-rootedness and beyond*, Handbook of enumerative combinatorics, Discrete Math. Appl. (Boca Raton), CRC Press, Boca Raton, FL, 2015, pp. 437–483. MR 3409348

[Cha97] Manoj K. Chari, *Two decompositions in topological combinatorics with applications to matroid complexes*, Trans. Amer. Math. Soc. **349** (1997), no. 10, 3925–3943. MR 1422892

[FMSV24] Luis Ferroni, Jacob P. Matherne, Matthew Stevens, and Lorenzo Vecchi, *Hilbert-Poincaré series of matroid Chow rings and intersection cohomology*, Adv. Math. **449** (2024), Paper No. 109733, 55. MR 4749982

[FS24] Luis Ferroni and Benjamin Schröter, *Valuative invariants for large classes of matroids*, J. Lond. Math. Soc. (2) **110** (2024), no. 3, Paper No. e12984. MR 4795885

[HRS21] Thomas Hameister, Sujit Rao, and Connor Simpson, *Chow rings of vector space matroids*, J. Comb. **12** (2021), no. 1, 55–83. MR 4195584

[Jef23] R. Amzi Jeffs, *Decompositions of augmented Bergman complexes*, Electron. J. Combin. **30** (2023), no. 1, Paper No. 1.6, 10. MR 4533712

[JKL18] Martina Juhnke-Kubitzke and Dinh Van Le, *Flawlessness of h-vectors of broken circuit complexes*, Int. Math. Res. Not. IMRN (2018), no. 5, 1347–1367. MR 3801465

[Lig97] Thomas M. Liggett, *Ultra logconcave sequences and negative dependence*, J. Combin. Theory Ser. A **79** (1997), no. 2, 315–325. MR 1462561

[NS04] Kathryn Nyman and Ed Swartz, *Inequalities for the h-vectors and flag h-vectors of geometric lattices*, Discrete Comput. Geom. **32** (2004), no. 4, 533–548. MR 2096747

[Oxl11] James Oxley, *Matroid theory*, second ed., Oxford Graduate Texts in Mathematics, vol. 21, Oxford University Press, Oxford, 2011. MR 2849819

[PRW08] Alex Postnikov, Victor Reiner, and Lauren Williams, *Faces of generalized permutohedra*, Doc. Math. **13** (2008), 207–273. MR 2520477

[Sta89] Richard P. Stanley, *Log-concave and unimodal sequences in algebra, combinatorics, and geometry*, Graph theory and its applications: East and West (Jinan, 1986), Ann. New York Acad. Sci., vol. 576, New York Acad. Sci., New York, 1989, pp. 500–535. MR 1110850

[Sta96] ———, *Combinatorics and commutative algebra*, second ed., Progress in Mathematics, vol. 41, Birkhäuser Boston, Inc., Boston, MA, 1996. MR 1453579

[Swa06] Ed Swartz, *g-elements, finite buildings and higher Cohen-Macaulay connectivity*, J. Combin. Theory Ser. A **113** (2006), no. 7, 1305–1320. MR 2259062